Featurization of Molecular Building Blocks for GRYFFIN

The electronic properties of all donor-bridge combinations (DBs) and 5452 prospective acceptors (As) were calculated using density functional theory (DFT) with the Orca software suite using the following functional, basis set, and solvent model: wB97x-D3/def2-SVP CPCM(Acetonitrile). Each building block had their reactive site (boron for DBs, halide for As) replaced with a hydrogen atom for all calculations. Each building block underwent ground state geometry optimization, followed by a ground state single point calculation and an excited state calculation using time dependent DFT (TDDFT) on the ground state optimized geometry (10 transitions, 5 expansion vectors in the iterative solution of the CI equations).

DFT features from ground-state calculations	
HOMO-1	Energy of the HOMO-1 (eV)
HOMO	Energy of the HOMO (eV)
LUMO	Energy of the LUMO (eV)
LUMO+1	Energy of the LUMO+1 (eV)
DipoleMoment	Magnitude of the ground state dipole moment (Debye)
Surface Charge	Surface charge calculated by the CPCM solvation model (elementary charges)
Charge Correction	Charge correction calculated by the CPCM solvation model (eV)
Free Energy	The change in free energy due to the CPCM solvation model (dispersion and cavity formation) (eV)
DFT features from excited-state calculations (TDDFT)	
PrimeExcite Strength	Strongest oscillator strength of the first 10 singlet transitions using transition electric dipole moments
PrimeExcite State	The singlet transition corresponding to the PrimeExcite Strength.
PrimeExcite Energy	Energy of the PrimeExcite State (eV)

Table S2: The DFT calculated descriptors of DBs and As

All building blocks were also featurized using structural and cheminformatics features extracted with RDKit.

Mass	Mass of the building block (amu)
HAcceptors	Number of hydrogen-bond acceptors
HDonors	Number of hydrogen-bond donors
LogP	Log(octanol-water partition coefficient)
Asphericity	Asphericity of the building block
Rg	The radius of gyration
TPSA	Topological Polar Surface Area
RingCount	Number of Rings
NumRotatableBonds	Number of Rotatable bonds
NumHeteroatoms	Number of Heteroatoms

Table S3: Structural Features of each building block

DFT features from excited state calculations	
S1 - S20	Energy of the first 20 singlet transitions from the ground state. This range of transitions extended up to 5.23 eV for all molecules. All states were used for DOS calculations. Only S1 – S10 were used for SVR models.
O1 - O20	Oscillator strength of the first 20 singlet transitions from the ground state, using the transition electric dipole moment. All transitions were used for DOS calculations. Only O1 – O10 were used for SVR models.
T1 - T20	Energy of the first 20 triplet states. This range of transition extended up to 4.66 eV for all molecules. All states were used for DOS calculations. Only T1 – T10 were used for SVR models.

Table S4: Additional DFT calculated descriptors of the DBAs (oligomers)

Using the features in **Table S4**, the following additional metrics were calculated in **Table S5**.

Calculated DFT-derived features	
Singlet DOS (SDOS)	The Density of States (DOS) of singlet transitions from TDDFT. Each state was broadened with a normalized gaussian, and the states were summed at a 0.1 eV resolution using the following equation, where E is the energy, n are the states, E_n is the energy of the nth state, and σ is the standard deviation of the gaussian (0.2 eV). $$DOS(E) = \sum_{n=1}^{20} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{E-E_n}{2\sigma^2}}$$
Weighted Singlet DOS (wSDOS)	The SDOS weighted by the transition electric dipole moment oscillator strength of the corresponding singlet transition from TDDFT. Each state was broadened with a gaussian with magnitude proportional to the oscillator strength of each transition, and the states were summed at a 0.1 eV resolution using the following equation, where O_n is the oscillator strength of the nth state and all other variables are the same as above. $$wDOS(E) = \sum_{n=1}^{20} \frac{O_n}{\sigma\sqrt{2\pi}} e^{-\frac{E-E_n}{2\sigma^2}}$$
Triplet DOS (TDOS)	The DOS of triplet transitions from TDDFT. Each state was broadened with a normalized gaussian, and the states were summed at a 0.1 eV resolution using the same equation as the Singlet DOS.

Table S5: Additional calculated metrics of the DBAs (oligomers)